Heurion 2.0: A Contract-Guarded Dual-Surface Architecture with Self-Evolving Memory for Citation-Grounded Clinical Research Authoring

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Abstract

Autonomous large language model (LLM) agents often edit documents via shell scripts and post-hoc file checks. In clinical research authoring, this workflow risks corrupting Office Open XML (`.docx`/`.pptx`) formatting, overwriting concurrent human edits, inserting unverified bibliographic IDs (DOIs/PMIDs), exposing Protected Health Information (PHI), and repeating guard failures across sessions.

We present **Heurion 2.0**, a clinical authoring platform combining a *Contract-Guarded Dual-Surface Architecture* with a *Three-Layer Self-Evolving Memory System*. A shared CRDT core (ProseMirror + Yjs) with stable block IDs projects a rich-text editor for clinicians and a guarded Annotated Markdown surface for AI agents with per-turn undo and original-slice OOXML export. Pre-write guards block unregistered citations, a two-stage PHI funnel pairs checksum/regex validators with a pre-masked local 1.5B model (`qwen2.5:1.5b`) and symbolic output filtering, and runtime guard/revert signals feed a utility-ranked procedural memory store isolated from declarative preferences. Across $1,286$ analysed live LLM turns on 10 PMC manuscripts and 680 guard test records, Heurion 2.0 achieved: (1) comparable completion (93.8% vs. 95.1%, $p = 1.00$), 0 vs. 4 content-corrupted untouched paragraphs ($p = 0.125$), and 100% (9/9 vs. 0/9, $p < 0.0001$) comment resolution in `.docx` editing (162 turns); (2) 98.6% (710/720) synthetic PHI recall (100% on 600 structured spans in Stage 1; 91.7% on 120 narrative spans in Stage 2) and 0/30 unconfirmed destructive tool executions (45 turns total); and (3) in a pre-specified 8-seed, 5-session ablation ($1,079$ turns), a directional 49% reduction in guard rejections on the primary mixed model (RR = 0.51, Holm $p = 0.1089$; NB2 sensitivity RR = 0.50, Holm $p = 0.0269$) and a 37% reduction in total execution signals (RR = 0.63, Holm $p = 0.0011$).

Keywords: Heurion 2.0, Collaborative AI Agents, Clinical Research Authoring, Yjs CRDT Co-Editing, Office Open XML (`.docx`/`.pptx`) Preservation, Model Context Protocol (MCP), Citation Hallucination Prevention, PHI Redaction, Self-Evolving Agent Memory, Human-AI Permission Symmetry

1 Introduction

Clinical and biomedical research workflows require high standards of traceability, structural fidelity, and regulatory compliance. A single clinical study typically spans patient longitudinal record curation, cohort screening, biostatistical modeling (e.g., baseline Table 1 generation, Kaplan–Meier survival curves, Cox proportional hazards regression, CONSORT flow diagrams), and the iterative authoring of peer-reviewed manuscripts (`.docx`) and academic presentation decks (`.pptx`) [1, 2]. While recent agentic systems have demonstrated strong capabilities in

code generation [3, 4] and open-domain literature synthesis [5, 6], deploying autonomous agents into clinical research environments exposes four architectural barriers:

1. **The Free-Form File Mutation Trap:** Conventional script-based document agents treat `.docx` and `.pptx` files as opaque disk artifacts manipulated via generated Python scripts (e.g., `python-docx`) followed by post-hoc quality checks. In practice, script-based rewrites can strip publisher XML attributes, lose paragraph addressing identifiers (`w14:paraId`) during LibreOffice layout passes, suffer from punctuation/whitespace reproduction drift in full-block replacements, and cannot selectively undo an AI turn once a human co-author makes concurrent edits.
2. **Citation Fabrication and Ungrounded Clinical Claims:** LLMs frequently hallucinate plausible-looking Digital Object Identifiers (DOIs), PubMed IDs (PMIDs), or numerical hazard ratios (HR) and p-values [9, 10, 11]. Post-hoc prompting alone cannot guarantee that every inline reference corresponds to a verified bibliographic record or that statistical assertions match the cited source text.
3. **Patient Privacy (PHI) Exposure and Permission Asymmetry:** Clinical investigators work with raw electronic medical records containing Protected Health Information (patient names, national ID numbers, admission codes, phone numbers) [12, 13]. Passing unredacted records to external LLMs conflicts with health-data protection law in both jurisdictions we target: China's Personal Information Protection Law (PIPL) classifies medical and health information as sensitive personal information that may be processed only for a specific purpose, with separate consent and strict protective measures, and imposes additional conditions on cross-border transfer [16]; the U.S. HIPAA Privacy Rule restricts disclosure of PHI unless it is de-identified, for example by removing the 18 Safe Harbor identifiers [15]. Conversely, granting agents unrestricted database or shell access creates privilege-escalation and indirect prompt-injection risks [17].
4. **Cross-Session Amnesia and Memory Pollution:** When an agent encounters a structural guard rejection (e.g., attempting to overwrite a human-edited block or inserting an unregistered citation) or when a clinician reverts an AI turn, stateless agents repeat the same operational failure in subsequent turns. Naively appending all historical observations into a single flat memory pool mixes subjective user writing preferences with hard operational constraints, exhausting context budgets with low-utility noise [19, 20].

1.1 Why Script-and-Audit File Mutation Fails in Collaborative Authoring

When general-purpose coding agents operate inside a workspace shell, they typically invoke Python scripts (`python-docx` or XML manipulators) to unpack, mutate, and repackage `.docx` archives on disk, optionally pairing edits with a post-turn audit script that checks text survival rates, paragraph drift, or DOI validity. Collaborative clinical authoring exposes three structural limitations in this "script-and-audit" paradigm:

1. **AI Editing Can Disrupt Document Structure:** When an agent edits a manuscript via ad-hoc scripts or whole-block re-serialization, it risks altering the underlying OOXML structure—stripping journal-template paragraph and run properties (`<w:pPr>`, `<w:rPr>`), breaking list numbering chains and multi-row table merges, detaching inline review comments and track-changes spans, and losing paragraph identifiers (`w14:paraId`) across layout passes.
2. **Risk of Overwriting Concurrent Human Edits:** Because script-based agents read a file snapshot, rewrite whole paragraphs or sections, and write the file back, any concurrent

edits made by the clinician mid-turn (such as inserting a clinical caveat in the same paragraph, adjusting a statistic, or attaching a comment) can be overwritten. Reconciling an AI-rewritten file with concurrent browser edits via post-hoc three-way merges is brittle.

3. **Post-Hoc Auditing Cannot Prevent Destructive Writes:** Auditing *after* the write occurs only after mutation: once a script has altered document formatting or overwritten a co-author's edit, a post-hoc check can only flag the discrepancy or trigger a coarse file-level rollback that discards the entire turn. Preventing structural corruption and human-edit loss requires replacing free-form file mutation with a shared, fine-grained collaborative data structure and pre-write structural guards.

1.2 The Heurion 2.0 Architecture and Contributions

Heurion 2.0 shifts from *post-hoc file auditing* to *pre-write structural contracts over a shared CRDT core*, coupled with a *signal-driven self-evolving execution harness*. Our primary contributions are fourfold:

- **Contract-Guarded Dual-Surface CRDT Engine (Section 3):** We design a unified ProseMirror + Yjs document core [7, 8] with persistent block addressing (`{#id}`), node-level optimistic concurrency control (`base_rev`), minimal-diff XML fragment updates that preserve inline marks and comment anchors, per-turn selective CRDT undo (`revertTurn`), and raw-slice OOXML patch export (`node_src`) that writes back original XML bytes for unmodified blocks.
- **Pre-Write Citation and Two-Stage Neuro-Symbolic PHI Guards (Section 4):** We enforce a syntactic pre-write ban on raw inline DOIs/PMIDs, requiring registration via Crossref, PubMed, and OpenAlex into atomic citation nodes (`[@c:id]`), and design a two-stage PHI redaction cascade that pairs sub-millisecond checksum/regex validators with a pre-masked local 1.5B small language model (`qwen2.5:1.5b`) and symbolic output filtering to redact both structured identifiers and unstructured clinical narratives within the local loopback boundary. For system completeness, we also outline our multi-agent claim verification pipeline and sandboxed biostatistical tools (which are not quantitatively evaluated in this study).
- **Three-Layer Self-Evolving Agent Architecture (Section 5):** We introduce a hierarchical adaptation loop comprising (Layer 1) side-effect-free pre-flight validation probes (`doc_validate` / `dry_run`), (Layer 2) a dual-track memory architecture that physically separates human-approved declarative preferences from runtime-crystallized procedural strategies ranked by a cue-based utility score under a fixed prompt budget, and (Layer 3) background meta-strategy distillation into dynamic agent profile overlays.
- **CI-Enforced Human-AI Permission Symmetry (Section 6):** We establish, and mechanically check via an automated parity test over the router table, an architectural invariant across all platform endpoints: every REST capability either maps 1-to-1 to an in-process Model Context Protocol (MCP) [18] tool executing under the user's exact RBAC identity, or is explicitly justified in a restricted exclusion set, with high-risk mutations gated by human-only confirmation cards.

2 System Overview and Design Principles

Figure 1 illustrates the end-to-end architecture of Heurion 2.0. The platform decouples *structured document/slide state mutation* from *sandboxed statistical computation*.

Heurion 2.0 is governed by three formal architectural invariants that enforce pre-write document integrity, human-AI permission symmetry, and memory orthogonality:

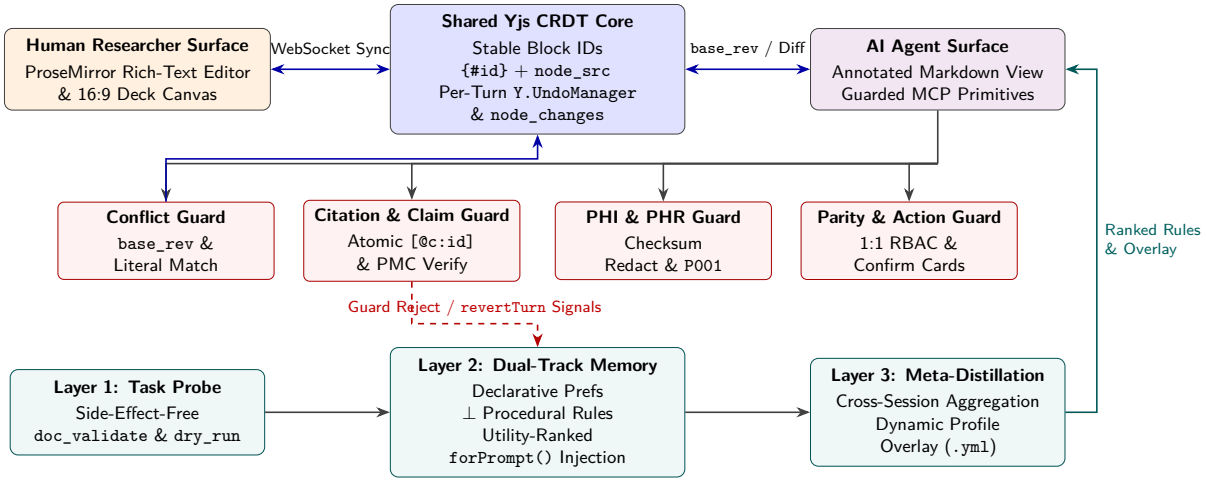


Figure 1: System architecture of Heurion 2.0, a contract-guarded clinical research and scientific authoring platform. Top: Human researchers and AI agents co-edit a shared Yjs CRDT document and slide core through synchronized surface projections (ProseMirror rich-text/deck canvas and Annotated Markdown MCP tools). Middle: All AI mutations pass through four deterministic pre-write structural guards (Conflict Guard, Citation & Claim Guard, PHI & PHR Guard, and Parity & Action Guard) before committing to the CRDT state. Bottom: Runtime guard rejections and human turn reverts feed a three-layer self-evolving memory system that dynamically injects high-utility procedural directives into future turns.

Architectural Invariant 1 (Pre-Write Structural Safety)**.** *Let S_r denote the CRDT document state at revision r, and let $\mathcal{O} = \{o_1, \ldots, o_k\}$ be a batch of structured edit operations submitted by an AI turn t against base revision $r_{base} \le r$. Let $\Delta_{user}(r_{base}, r)$ be the set of block IDs modified by human actors in $(r_{base}, r]$, $\mathcal{P}_{sugg}(S_r)$ be the set of blocks holding unresolved clinician track-changes suggestions, $Anchors(b)$ be the set of open review-comment anchors attached to block b, and $\mathcal{R}_{cite}(S_r)$ be the set of verified citation identifiers registered in the document store. The state transition $\mathcal{T}(S_r, \mathcal{O}, r_{base})$ commits if and only if every $o_i \in \mathcal{O}$ with target block $b = target(o_i)$ satisfies all four pre-write conditions:*

$$\begin{aligned} BlockMut(o_i) &\implies b \notin \Delta_{user}(r_{base}, r) \cup \mathcal{P}_{sugg}(S_r), \\ ReplaceText(o_i) &\implies \big|Matches\big(find(o_i),\, S_r[b]\big)\big| = 1 \;\wedge\; b \notin \mathcal{P}_{sugg}(S_r), \\ Anchors(S_r[b]) &\subseteq Anchors\big(o_i(S_r[b])\big) \cup DropAnchors(o_i), \\ Cites(o_i) &\subseteq \mathcal{R}_{cite}(S_r) \;\wedge\; RawCitePattern(o_i) = \emptyset. \end{aligned} \tag{1}$$

Otherwise, the entire batch $\mathcal{O}$ is atomically rejected prior to mutating S_r, returning a structured diagnostic code (`conflict_user_edited`, `pending_suggestion`, `text_not_found`, `ambiguous_match`, `anchor_has_open_comments`, or `citation_not_registered`) and the current block snapshot.

Architectural Invariant 2 (Human-AI Permission Symmetry)**.** *Let $\mathcal{E}_{REST}$ be the set of all HTTP API routes exposed to the web user interface, and let $\mathcal{E}_{MCP}$ be the set of Model Context Protocol (MCP) tools exposed to the AI agent. Every route $e \in \mathcal{E}_{REST}$ satisfies the disjoint partition:*

$$\mathcal{E}_{REST} = \phi(\mathcal{E}_{MCP}) \sqcup \mathcal{E}_{NOT_FOR_AI}, \tag{2}$$

where $\phi : \mathcal{E}_{MCP} \to \mathcal{E}_{REST}$ is an in-process Loopback dispatcher that executes under the authenticated user's exact session token and tenant isolation context, and $\mathcal{E}_{NOT_FOR_AI}$ is an explicitly audited set of human-only lifecycle endpoints (e.g., credentials, high-risk confirmation approval).

Architectural Invariant 3 (Declarative-Procedural Memory Orthogonality)**.** *Let $\mathcal{M}_D$ denote the declarative memory store (subjective authoring preferences, stylistic conventions, domain terminology) and $\mathcal{M}_P$ denote the procedural harness store (operational guard avoidance rules and subagent delegation policies). Updates to $\mathcal{M}_D$ require explicit human confirmation (status $\in \{\texttt{candidate} \xrightarrow{human} \texttt{confirmed}\}$), whereas $\mathcal{M}_P$ evolves autonomously from deterministic runtime execution signals Σ_{exec} without polluting $\mathcal{M}_D$ or requiring manual administrative curation.*

3 Dual-Surface Co-Editing and Original-Slice Document Engine

How can autonomous AI agents edit complex Word (`.docx`) manuscripts and PowerPoint (`.pptx`) presentation decks without corrupting journal-template formatting or overwriting concurrent human edits? A central architectural observation in Heurion 2.0 is that Office Open XML (`.docx` and `.pptx`) files are delivery formats, not collaborative editing data structures. When an agent edits a `.docx` file on disk via a Python script while a human views the document in a browser, the two actors operate on divergent state representations. Heurion 2.0 eliminates this divergence by replacing direct file scripting with a Contract-Guarded Dual-Surface CRDT Engine [7, 8].

3.1 Shared CRDT Core with Lifecycle-Stable Block Addressing

In Heurion 2.0, every manuscript and presentation deck resides in a server-authoritative Yjs CRDT document (`Y.Doc`) backed by SQLite state snapshots and an append-only revision log (`doc_commits`). Every block-level ProseMirror node (heading, paragraph, list item, table, figure, or slide element) is assigned a short, lifecycle-stable identifier (`attrs.id`, rendered as `{#id}`). When a human splits a paragraph or pastes content in the browser, the server-side `flush()` handler automatically detects duplicate or missing block IDs and assigns fresh unique IDs inside a system CRDT transaction, ensuring the addressing space remains bijective.

From this single CRDT core, Heurion 2.0 projects two synchronized surfaces:

1. **Human Surface:** A real-time collaborative ProseMirror rich-text editor (for manuscripts) and an interactive 16:9 SVG/HTML slide canvas (for decks) synchronized via WebSocket (`y-protocols/sync`).
2. **Agent Surface:** A compact *Annotated Markdown* view generated on demand via `doc_read` and `slide_read`. Each block is suffixed with its stable address `{#id}`, inline citations are rendered as atomic tokens `[@c:cite_id]`, figures as `![alt](asset:id)`, and active inline review comments as `{#c:comment_id}`.

3.2 Node-Level Conflict Guards and Minimal-Diff Fragment Updates

To prevent an AI turn in Heurion 2.0 from overwriting concurrent human edits, every `doc_edit` call specifies the `base_rev` read by the agent. As formalized in Invariant 1, the platform queries the `node_changes` index for all blocks modified by human actors since `base_rev`:

- **Block-Level Structural Operations (`replace_block`, `delete`):** If a target block $b \in \Delta_{\text{user}}(r_{\text{base}}, r)$, the operation is rejected with error code `conflict_user_edited`, returning the human's latest block markdown so the agent can adapt. Similarly, blocks carrying unresolved track-changes suggestions ($b \in \mathcal{P}_{\text{sugg}}(S_r)$) are protected by the `pending_suggestion` guard, and blocks holding open review comments require either preserving the comment anchors or explicitly declaring `drop_anchors` (`anchor_has_open_comments`).

- **Substring Replacement (`replace_text`):** For surgical phrase edits within a paragraph, the literal `find` substring acts as its own optimistic concurrency guard: if the human edited a different sentence in the same paragraph, `find` still matches uniquely and succeeds; if the human altered the target phrase itself or if `find` matches multiple occurrences, validation fails cleanly with `text_not_found` or `ambiguous_match`.
- **Mark-Preserving Minimal Diffs:** Notably, `replace_text` does not re-parse the paragraph from plain text (which would destroy inline bold/italic spans, `[@c:id]` citation nodes, and comment highlights outside the matched range). Instead, it locates the character offsets $[s, e)$ in the ProseMirror node's flattened text stream and replaces only that slice, after which `writeFragment` updates only the modified XML children between the common prefix and suffix in the underlying `Y.XmlFragment`.

3.3 Per-Turn Selective Undo in CRDT Space (`revertTurn`)

In collaborative scientific writing, a researcher may ask the AI to revise three sections, continue manually editing Section 2, and subsequently decide to undo the AI's changes. Global linear undo (`Ctrl+Z` or version rollback) would destroy the researcher's manual edits in Section 2.

Heurion 2.0 solves this via a two-tier *Per-Turn Selective Undo* mechanism (`revertTurn`):

1. **In-Memory CRDT Origin Tracking:** When an AI turn t commits changes to document d, the runtime binds a dedicated `ServerOrigin` object $o_t = \{\texttt{server} : \text{true}, \texttt{actor} : \texttt{'ai'}, \texttt{turnId} : t\}$ and attaches a scoped `Y.UndoManager` configured with `trackedOrigins` $= \{o_t\}$. Invoking `revertTurn(d, t)` executes `um.undo()` solely on the CRDT items inserted or deleted under origin o_t, leaving concurrent human character insertions—even within the same paragraph—intact.
2. **Persistent Node-Level Fallback across Server Restarts:** Before turn t's first write, the runtime persists the pre-turn Yjs state vector in `turn_doc_bases`. If the server restarts before `revertTurn` is invoked, the engine reconstructs the pre-turn document tree, computes the per-block diff against turn t's recorded `node_changes`, reverts all AI-modified blocks untouched by the user since $r_{\min}(t)$, and safely skips blocks subsequently edited by the user (returning their IDs in `skipped`).

3.4 Original-Slice OOXML Patch Export

Clinical manuscripts are frequently imported from journal-specific Word templates containing custom styles, headers/footers, macros, and embedded vector objects. During `.docx` or `.pptx` import, Heurion 2.0 stores the original archive in `doc_packages` and records the raw XML slice (`<w:p>` or `<w:tbl>`) of every imported block in `node_src` keyed by block ID. Upon export (`docx-export.ts`, `pptx-export.ts`), any block whose ProseMirror node is structurally equal (`node.eq(baselineNode)`, meaning its text, inline formatting marks, and attached comment range markers are unchanged) to its imported baseline is written back using its exact original `node_src` XML bytes; only blocks whose text or inline comment/mark structure was mutated during the session are re-serialized, preserving paragraph properties (`<w:pPr>`), run fonts, and paragraph identifiers (`w14:paraId`) from the original template.

4 Citation-Grounded Evidence Loop and Privacy-Preserving Clinical Pipeline

How can a clinical authoring platform prevent LLM citation fabrication while protecting patient privacy (PHI) in medical records? Heurion 2.0 addresses both requirements through deterministic pre-write syntactic guards and a two-stage neuro-symbolic PHI redaction funnel.

4.1 Syntactic Citation Guard and Atomic Citation Nodes

To prevent fabricated references from entering manuscript text, Heurion 2.0 blocks agents from writing raw bibliographic identifiers or hand-numbered brackets in prose. During pre-write validation in `OpService`, every candidate text string is scanned against structural regular expressions detecting bare DOIs (`10.\d{4,9}/…`), bare PMIDs (`PMID:\s*\d+`), or unlinked reference markers. Any violation triggers `citation_not_registered` before modifying the document.

Instead, the agent must invoke `insert_citation`, which queries live metadata APIs (Crossref, NCBI PubMed E-utilities, and OpenAlex), verifies the record's existence, stores the canonical metadata (authors, title, journal, year, volume, pages, DOI/PMID) in the document's `citations` table, and returns a registered token `[@c:cite_id]`. In ProseMirror, `[@c:cite_id]` is parsed into an atomic inline node (`atom: true`) that dynamically numbers itself according to citation order and automatically compiles the terminal References section.

4.2 System Context: Claim Verification and Biostatistical Sandbox

In addition to the pre-write citation guard evaluated in Section 7, the platform includes a multi-agent claim verification service (`ClaimService.verify`) and a sandboxed biostatistical execution toolchain (included here for architectural context; clinician-graded evaluation of these modules is left to future work). Briefly, `verify_claims` segments manuscript blocks into sentences, retrieves PubMed abstracts and available PubMed Central (PMC) full texts for cited `[@c:cite_id]` records, and classifies quantitative assertions into four verdicts (`supported`, `contradicted`, `unsupported`, or `uncited_stat`), attaching problematic claims as inline ProseMirror review comments. For statistical analysis, computation is delegated to isolated Python/R tools (`biostatistician` subagent) rather than free-text LLM arithmetic.

4.3 Multi-Tenant PHI Redaction Funnel

To bridge clinical data management and manuscript authoring without exposing patient identities [12, 13, 14], Heurion 2.0 enforces four privacy safeguards:

- **Per-Tenant Database Isolation & Envelope Encryption:** Each organization tenant operates on a physically isolated SQLite database file (`tenants/<org_id>/platform.db`) with per-tenant Data Encryption Keys (DEKs) wrapped by a master Key Encryption Key (KEK).
- **Two-Stage Neuro-Symbolic PHI Funnel (`phi-scan.ts`):** Using an external cloud LLM to inspect raw clinical text for PHI creates a privacy paradox: sending unredacted patient records to a third-party API in order to redact them already violates tenant data-residency boundaries. Conversely, running a frontier-scale 70B+ parameter model on-premise imposes high GPU requirements for clinical research teams. Heurion 2.0 addresses this trade-off through a task-decomposed, two-stage neuro-symbolic funnel that operates within the local environment:
  1. *Stage 1 — Sub-Millisecond Deterministic Checksum & Subtype Rules (`scanPhi`/`redactPhi`):* Deterministic rules execute first (0.011 ms per note) to validate and mask mathematically structured or field-labeled identifiers: 18-digit national IDs (ISO 7064 MOD 11-2 checksum + calendar bounds), settlement/bank cards (ISO/IEC 7812 industry prefix + Luhn mod-10 checksum with DOI/URL boundary exclusion), 11-digit mobile numbers, medical record numbers, and co-occurring patient-name + clinical diagnosis pairs spanning both pure-CJK and mixed alphanumeric/Roman-numeral subtypes (e.g., Type-2 diabetic nephropathy, diffuse large B-cell lymphoma).
  2. *Stage 2 — Pre-Masked Local Small Language Model with Symbolic Output Guard (`createLocalSlmPhiDetector` + `redactPhiCascaded`):* Because Stage 1 pre-masks structured

identifiers before Stage 2 is invoked, the Stage-2 model does not receive raw national IDs, phone numbers, or bank cards, and does not need to perform arithmetic checksum verification. Instead, a compact 1.5B-parameter local small language model (`qwen2.5:1.5b`, 986 MB, served via local Ollama and gated by `assertLocalPhiEndpoint` to allow only loopback `127.0.0.1`/`::1` or RFC 1918 private intranet hosts) extracts open-vocabulary residual spans in free-form ward-round narratives: unlabeled person/relative names (`unlabeled_person_name`) and specific residential/workplace addresses (`address_or_org`). Finally, a deterministic post-validator (`normalizeAndValidateStage2Entity`) verifies exact substring presence in the source text, strips kinship/location prepositions, checks Chinese family-surname membership and medical-suffix exclusions for names, and requires spatial/organizational markers for addresses—filtering out small-model hallucinations and clinical-term false positives.

- **Pseudonymous Subject Mapping & PHR Red-Lines:** Within research cohorts, real patient identifiers are replaced by pseudonymous subject codes (`P001`, `P002`, …) via an in-memory translation table. Furthermore, a dedicated Personal Health Record guard (`phr_guard`) enforces read-only access on patient-uploaded primary medical records, preventing AI tools from altering ground-truth clinical evidence.
- **Regulatory Scope:** These mechanisms reduce exposure but do not by themselves establish compliance with either PIPL or HIPAA. Pseudonymous subject codes are de-identification rather than anonymization: under PIPL, data that can be re-linked with a retained mapping table remains personal information [16]. The current scanner implements mainland Chinese identifier formats (resident ID numbers, mobile numbers, hospital record numbers, Chinese surnames/addresses); it does not yet cover HIPAA Safe Harbor identifiers such as Social Security numbers, U.S.-format telephone numbers, full dates, or English geographic subdivisions [15], so U.S. deployment requires extending it.

5 Three-Layer Self-Evolving Agent Architecture

A recurring challenge in contract-guarded agent environments is helping the LLM internalize operational boundaries without manual prompt engineering. When an agent's tool call is blocked by a pre-write guard or reverted by a user, how does the system help the agent adapt both within the current task and across future sessions?

As shown in the bottom tier of Figure 1, Heurion 2.0 implements a three-layer adaptation architecture operating across three timescales:

5.1 Layer 1: Task-Level Pre-Flight Probe (`doc_validate` & `dry_run`)

At the intra-turn millisecond timescale, agents in Heurion 2.0 can invoke `doc_validate` (or pass `dry_run: true` to `doc_edit`) prior to committing complex multi-block edits. The probe executes the full suite of conflict guards, citation registration checks, anchor integrity checks, and post-edit structural diffs against an ephemeral in-memory clone of the ProseMirror document tree without incrementing `rev`, mutating the Yjs CRDT state, or polluting the undo stack. If a violation is detected, the probe returns `{valid: false, code, hint, current}`, enabling zero-cost self-correction within the same reasoning turn.

5.2 Layer 2: Session-Level Dual-Track Memory and Utility-Ranked Injection

At the cross-turn session timescale, Heurion 2.0 addresses memory pollution by enforcing **Declarative-Procedural Orthogonality** (Invariant 3):

- **Track A — Declarative Memory (`MemoryService`):** Captures domain conventions and author preferences (e.g., "Always report continuous skewed variables as median (IQR)", "Use British English spelling"). Because stylistic preferences are subjective, newly extracted items enter a `candidate` state and require explicit human approval in the UI before becoming `confirmed`.
- **Track B — Procedural Harness Memory (`ProceduralHarnessService`):** Captures objective operational rules for navigating guards, tools, and specialist subagents. Whenever `OpService` rejects an AI edit or a user invokes `revertTurn`, the `ExecutionSignals` collector automatically records a structured runtime signal $s = (\text{owner}, \text{doc_id}, \text{turn_id}, \text{kind}, \text{code}, \text{detail})$ into the `execution_signals` table.

At the end of an authoring session, `evolveFromSignals()` clusters unconsumed execution signals by $(\text{kind}, \text{code})$. Any failure pattern occurring at least $\tau = 2$ times either increments the empirical evidence counter e_r of an existing procedural rule $r \in \mathcal{M}_P$ or crystallizes a new active procedural directive—requiring zero manual UI intervention.

Context-Aware Utility Ranking. Injecting all procedural rules into every turn would waste context tokens. Before each turn on document d with user prompt q, `ProceduralHarnessService.forPrompt()` computes a relevance score $U(r \mid d, q)$ for each active rule $r \in \mathcal{M}_P$. Guard codes never appear in a user's request, so a rule is not matched against q by its code; instead each guard code is associated with a *message cue* (keywords in q, e.g., DOI/PMID or citation terms for `citation_not_registered`, replacement verbs for `ambiguous_match`) and/or a *document-state cue* computed from d (human edits within the last two revisions for `conflict_user_edited`, open comment threads for `anchor_has_open_comments`, pending suggestions for `pending_suggestion`, the presence of a table for table-edit codes):

$$U(r \mid d, q) = \alpha \cdot e_r + \beta \cdot \mathbb{I}(\text{msgCue}_r(q)) + \gamma \cdot \mathbb{I}(\text{stateCue}_r(d)) \,, \tag{3}$$

where e_r is the empirical evidence count, $\mathbb{I}(\cdot)$ is the indicator function, and $(\alpha, \beta, \gamma) = (0.1, 5, 8)$. Baseline subagent-delegation rules, whose trigger is a keyword pattern rather than a guard code, use that pattern as their message cue; codes without a defined cue are ranked by e_r alone. These weights are hand-set heuristics, not tuned values. Rules are sorted in descending order of $U(r \mid d, q)$ and greedily packed: a rule of length ℓ is charged $\ell + 20$ characters and skipped if it would exceed the budget B (default $B = 1000$), up to at most $K = 4$ rules.

5.3 Layer 3: System-Level Meta-Strategy Distillation (`HarnessEvolutionService`)

At the multi-session platform timescale, Heurion 2.0's `HarnessEvolutionService` periodically aggregates system-wide execution signals and active procedural rules. When cumulative unanalyzed signals exceed a threshold, a background meta-evolver distills recurring failure root causes into concise system-level behavioral directives (2–5 rules, ≤ 100 characters each) and serializes them into a dynamic YAML profile overlay (`evolved.cordis.yml`):

```
# Auto-generated by Heurion Self-Evolution Engine
- id: system-prompt
  config:
    personaSuffix: >-
      [Evolved Self-Improvement Execution Directives]:
      - Run doc_validate prior to mutating blocks with recent user activity
      - Register all DOI/PMID entries via insert_citation before inline referencing
```

When `HarnessPool` spawns or refreshes an agent runtime, `getPatches()` transparently mounts `evolved.cordis.yml` alongside the base platform profile, updating the agent's system persona without user disruption.

6 CI-Enforced Human-AI Permission Symmetry

A common security and usability gap in enterprise AI applications is the disconnect between what a human user can perform via the graphical user interface and what the AI agent can perform via tool calls. Often, developers either grant the agent direct database access (bypassing business-layer authorization) or expose only a subset of tools.

Heurion 2.0 enforces **Human-AI Permission Symmetry** (Invariant 2) across all platform capabilities:

1. **In-Process Loopback Execution under User Identity:** Every administrative and clinical MCP tool [18] (`patient_*`, `study_*`, `doc_manage`, `kb_manage`, `dataset_manage`, `org_manage`) executes by invoking `invokeHttp(api, userId, method, path, body)` directly against the Hono HTTP router in-process. The call carries a short-lived internal JWT minted for `userId` and tagged with header `X-Heurion-Via: ai`. Consequently, the agent inherits the exact same multi-tenant isolation, study role checks (PI, co-investigator, statistician, reviewer), and audit logging as the human user.
2. **Human-Only Confirmation Cards for High-Risk Actions:** When an AI tool invokes a destructive or irreversible endpoint (e.g., purging a document, freezing a clinical study cohort, or removing a team member), the HTTP middleware intercepts the `X-Heurion-Via: ai` header and—rather than executing the mutation—creates a `pending_actions` record and renders an interactive confirmation card in the user's chat stream. Only when the human clicks *Confirm* in the browser does the UI re-issue the request with `X-Heurion-Via: ai-confirmed`, completing execution.
3. **Automated CI Parity Verification (`parity.test.ts`):** Our test suite introspects the Hono application router table at build time and asserts that every registered `/api/*` endpoint appears either in the MCP parity mapping table ($\phi(\mathcal{E}_{\text{MCP}})$) or in the documented $\mathcal{E}_{\text{NOT_FOR_AI}}$ exclusion registry. Any newly added API route missing an MCP counterpart or explicit exemption immediately fails continuous integration.

7 Empirical Evaluation

We evaluate Heurion 2.0 through a two-tier empirical protocol combining:

1. **Deterministic and Local-SLM Benchmarks (`run_eval.ts`):** 680 test records against the platform's TypeScript/SQLite/Yjs engine and local Ollama runtime, comprising 50 CRDT concurrency/revert trials (`exp1`) and 630 citation and PHI guard records (`exp2b`: 100 bare-citation sentences, 200 synthetic clinical notes containing 720 positive PHI spans, and 330 negative controls), plus a 60-trial deterministic memory-budget simulation (`exp3_self_evolution`). Local Stage-2 PHI extraction used `qwen2.5:1.5b` (986 MB, Ollama v0.20.5 on `http://127.0.0.1:11434/api/generate`, `temperature=0`, `format='json'`).
2. **Pre-Specified Live LLM Evaluation (`ANALYSIS_PLAN.md`):** 1,286 analysed live agent turns (1,287 executed: 162 in `exp2a-papers`, 45 in `exp2c`, and 1,079 analysed of 1,080 in `exp3`) over 10 open-access (CC BY) clinical research manuscripts from PubMed Central (PMC; 7 training and 3 held-out manuscripts spanning randomized controlled trials, cohort studies, and meta-analyses). All online turns were executed on October 19, 2026 using model ID `deepseek-flash` via `@deepseek-ai/dsh-sdk-client` against

Table 1: Architectural mechanism comparison across direct script mutation (`python-docx`), script mutation with post-hoc file auditing, and Heurion 2.0's contract-guarded dual-surface CRDT architecture. Entries describe system design mechanisms, not measured rates.

Dimension	Direct Script (`python-docx`)	Script + Post-Hoc Audit	Heurion 2.0 (Ours)
Source of Truth	Raw `.docx` on Disk	Raw `.docx` + Derived Projection	**Shared Yjs CRDT + SQLite**
Block Addressing	None (Regex/XPath)	File-Native `w14:paraId` / Shape ID	**Server-Assigned `{#id}`**
Export of Untouched Blocks	Full Re-serialization	Whole-File Save	**Original XML Slice (`node_src`)**
Concurrent Human Edits	Silent Overwrite	Post-Hoc Node-Level 3-Way Merge	**Pre-Write `base_rev` Guard**
Undo of One AI Turn	Unsupported	File-Level Rollback	**Per-Turn `Y.UndoManager`**
Unregistered Citations	Not Checked	Post-Write Audit	**Pre-Write Block**
Pre-Flight Validation	None	Subprocess Audit Script	**In-Memory Dry Run**

Table 2: Heurion 2.0 dual-surface CRDT engine micro-benchmark results (`exp1`) over $N = 50$ trials on the synthetic clinical protocol template. Latencies are wall-clock means of in-process calls on a single developer machine.

Metric	Result
Untouched blocks reproduced verbatim on export (heading + final paragraph)	50/50
Stale `replace_block` rejected with `conflict_user_edited`	50/50
Turn revert undoes AI edits while keeping the human edit in the same paragraph	50/50
Mean pre-flight validation latency (`OpService.validate`)	0.168 ms
Mean guarded edit latency (`OpService.edit`, two ops)	0.351 ms
Mean per-turn revert latency (`revertTurn`)	0.128 ms

`https://api.deepseek.com/anthropic/v1` with default SDK sampling parameters. We selected a fast, compact model (`deepseek-flash`) because lightweight models exhibit higher baseline rates of boundary, anchor, and citation-formatting violations than frontier reasoning models, providing a stress test for pre-write structural guards and procedural memory crystallization. Across all live experiments, each turn spawns a fresh agent session (`HARNESS_FRESH_SESSION=1`) with declarative memory auto-extraction and Layer 3 background overlays disabled (`HARNESS_EVOLVE=0`) so that simulated user trajectories (parameterized by random seeds) remain statistically independent.

7.1 Experiment 1: Dual-Surface CRDT Engine vs. Script-Based File Mutation

Table 1 contrasts the architectural mechanisms of direct script mutation (`python-docx`), script mutation paired with post-hoc file auditing, and Heurion 2.0.

Experiment 1a: Deterministic Engine Micro-Benchmarks (`exp1`). Table 2 reports in-process micro-benchmarks of Heurion 2.0 over $N = 50$ trials on a six-block `.docx` clinical protocol template (one styled heading, a bold paragraph, an italic paragraph, two list items, and a plain paragraph). In each trial, an AI turn applies two `replace_text` edits (p_1, p_2), a simulated human edits a different phrase inside p_2, a stale `replace_block` is probed, and the AI turn is selectively reverted via `revertTurn`. All 50/50 trials reproduced the original `<w:p>` XML bytes of the untouched heading and final paragraph verbatim via `node_src`, rejected the stale overwrite with `conflict_user_edited`, and undid the AI edits while preserving the human's intra-paragraph edit in p_2 at sub-millisecond latency (0.168 ms validation, 0.351 ms edit, 0.128 ms revert).

Experiment 1b: End-to-End Clinical Manuscript Editing—Heurion 2.0 MCP vs. `python-docx` (`exp2a-papers`). To test how the two editing surfaces behave under

Table 3: End-to-end clinical manuscript editing comparison (`exp2a-papers`, 162 live LLM turns across 81 paired tasks per arm, 3 simulated user seeds × 3 manuscripts × 9 archetypes) between Heurion 2.0 MCP primitives and direct `python-docx` file scripting. 95% confidence intervals use the Wilson score method; p-values are two-sided Fisher's exact tests. †In 2 `open_comment_anchor` turns (`u3-s3`, `u1-s3`), the MCP agent invoked `delegate_task` (`fact_checker`) without being asked to, which attached 7 unrequested verification comments (`<w:commentRangeStart/>`) to non-target paragraphs; their text was unchanged (`content_changed = 0`), but these are AI-initiated writes outside the task scope. ‡In full-paragraph rewrites, the MCP agent passed `drop_anchors: true` (6/9 cases) to clear obsolete range tags after replying to all 9/9 review threads via `comment_reply`, whereas `python-docx` run-level edits left the sibling `<w:commentRangeStart/>` tags in place without replying to `word/comments.xml`.

Evaluation Endpoint	Heurion 2.0 MCP ($n = 81$)	Direct `python-docx` ($n = 81$)	Fisher p
Task completion rate across all 9 archetypes	76/81 (93.8%, CI 86.4–97.3%)	77/81 (95.1%, CI 88.0–98.1%)	1.0000
Untouched paragraphs with unintended content changes	0/32,995 (0.00%)	4/32,995 (0.01%)	0.1250
Untouched paragraphs preserved byte-identically†	32,988/32,995 (99.98%)	32,991/32,995 (99.99%)	0.5488
Edited paragraphs preserving native `w14:paraId`	41/45 (91.1%, CI 79.3–96.5%)	37/45 (82.2%, CI 68.7–90.7%)	0.3529
Edited paragraphs preserving layout, fonts, and emphasis	44/45 (97.8%)	45/45 (100.0%)	1.0000
Review comments read & replied in thread (`acknowledged`)	**9/9 (100.0%, CI 70.1–100%)**	0/9 (0.0%, CI 0.0–29.9%)	**< 0.0001**
Inline `<w:commentRangeStart/>` tag retained (6/9 cleared via `drop_anchors`)‡	3/9 (33.3%, CI 12.1–64.6%)	**9/9 (100.0%, CI 70.1–100%)**	**0.0090**
Clinician pending track-change suggestion preserved	9/9 (100.0%, CI 70.1–100%)	9/9 (100.0%, CI 70.1–100%)	1.0000
Unregistered bare DOI / PMID injected into manuscript body	0/81 (0.0%, CI 0.0–4.5%)	1/81 (1.2%, CI 0.2–6.7%)	1.0000

live LLM tool use on full-length journal manuscripts, we compared Heurion 2.0's MCP surface (`PROCEDURAL_MODE=off`) against a baseline coding agent using the same model (`deepseek-flash`) and identical authoring guidelines (`baseline.cordis.yml`) to edit `manuscript.docx` directly in a workspace directory. In the baseline arm, the agent runs inside the same `DeepSeekHarness` runtime with unrestricted `bash` and `python3` access (including `python-docx`, `zipfile`, and `xml.etree.ElementTree`) and is explicitly instructed to *"preserve the existing formatting, comments and tracked changes"*. Both arms executed 3 paired simulated user seeds (seeds 201–203) × 3 PMC clinical manuscripts × 9 authoring task archetypes (`concurrent_edit`, `citation`, `pending_suggestion`, `open_comment_anchor`, `term_unify`, `table_row`, `out_of_scope_revert`, `p_value_format`, and `move_subsection`), yielding 81 turns per arm (162 turns total; 32,995 untouched paragraphs per arm). Both arms exported standard `.docx` archives evaluated by the same OOXML inspector (`manuscripts/compare.ts`).

Table 3 summarizes three empirical observations from `exp2a-papers`:

1. **Comparable Task Completion and Structural Preservation Rates:** Task completion did not differ significantly between Heurion 2.0 MCP (76/81 = 93.8%) and direct `python-docx` scripting (77/81 = 95.1%, Fisher $p = 1.0000$). Across 32,995 non-target paragraphs, Heurion 2.0 produced 0 unintended text modifications compared to 4 altered non-target paragraphs in the `python-docx` arm ($p = 0.1250$), and retained `w14:paraId` on 41/45 (91.1%) vs. 37/45 (82.2%, $p = 0.3529$) edited paragraphs; neither difference reached statistical significance at $n = 81$ turns per arm. Why were 7 of 32,995 untouched paragraphs in Heurion 2.0 not byte-identical (32,988/32,995 = 99.98% vs. 32,991/32,995 = 99.99%)? Inspection of the turn logs showed that all 7 paragraphs occurred in two `open_comment_anchor` turns (`u3-s3` on `PMC8861838`, 4 paragraphs; `u1-s3` on `PMC10260499`, 3 paragraphs) where the MCP agent voluntarily invoked the `fact_checker` subagent (`delegate_task`), which attached unrequested `<w:commentRangeStart/>` verification comments to those 7 paragraphs while leaving their text unchanged (`content_changed = 0`). We therefore report these as out-of-scope side effects: Heurion 2.0 made 0 unintended text changes but 7 unrequested comment insertions on non-target paragraphs.
2. **Comment-Thread Resolution vs. Unassisted File Scripting:** In the `open_comment_anchor` task (which instructs the agent to rewrite a Methods paragraph according to two

attached clinician review comments), although the baseline agent had full Python/XML access to `word/comments.xml`, standard `python-docx` scripts written by the agent iterated over `Document.paragraphs` (which does not expose `word/comments.xml`) and lacked a dedicated comment-thread tool or pre-write guard; consequently, the file agent modified the paragraph's text runs while leaving the sibling `<w:commentRangeStart/>` XML tags in place (9/9 vs. 3/9, $p = 0.0090$) and never appended a reply to `word/comments.xml` (0/9 acknowledged). Conversely, Heurion 2.0's `anchor_has_open_comments` guard and `comments_list`/`comment_reply` tools prompted the MCP agent to inspect the comments, declare `drop_anchors` when replacing the full paragraph (6/9 cases), and post structured replies in all 9/9 (100%, $p < 0.0001$) review threads.

3. **Methodological Note on File-Arm Concurrent Edits:** Both arms preserved 9/9 clinician insertions in `concurrent_edit`. Inspection of the event logs revealed that because the file agent executes a single atomic `bash` call (`Document('manuscript.docx') ... save()`) whereas the harness callback `humanEditsFile` fires upon `tool_result` completion, the simulated clinician insertion in the file arm occurred between separate `bash` invocations rather than mid-flight between an in-memory read and write. Mid-flight stale-snapshot overwrites are evaluated deterministically in Table 2 (50/50 intercepted) and in Experiment 3.

7.2 Experiment 2: Pre-Write Citation, PHI, and High-Risk Action Guards

Experiment 2a: Two-Stage Deterministic + Local SLM PHI and Citation Guards (`exp2b`). To quantify the complementary roles of Stage-1 deterministic validators (`scanPhi`/`redactPhi`) and the Stage-2 local small language model (`qwen2.5:1.5b` via loopback Ollama, `redactPhiCascaded`), we evaluated Heurion 2.0 across 630 test records comprising 100 citation-guard sentences, 200 synthetic positive clinical notes containing 720 PHI spans across 7 categories, and 330 negative controls across 3 strata (Table 4):

1. **Unregistered Inline Citations ($N = 100$ distinct sentences):** 40 bare DOIs across 8 biomedical publisher prefixes, 30 `https://doi.org/` URLs, and 30 bare PMIDs. All 100/100 (100%) were intercepted prior to CRDT mutation with `citation_not_registered`.

2. **Stage-1 vs. Stage-2 Local SLM Recall across 720 Synthetic PHI Spans ($N = 200$ notes):** Each synthetic note contained 3 structured/labeled identifiers (120 spans per category across 5 categories = 600 spans), and 60 notes additionally included unstructured ward-round narrative clauses containing an unlabeled relative/caregiver name (`unlabeled_person_name`, 60 spans) and a residential street/building/room address (`address_or_org`, 60 spans). As shown in Table 4, **Stage 1 alone** (0.011 ms/note) achieved 100.0% (600/600) recall across the five structured and labeled categories—including 120/120 on mixed alphanumeric/Roman-numeral diagnosis subtypes (where a pure-CJK diagnosis regex recalled 90/120 = 75.0%)—but recalled 0/120 (0.0%) on unlabeled free-narrative names and residential addresses (600/720 = 83.3% overall). Because the 200 synthetic notes were generated from deterministic templates authored by the same team that implemented Stage 1, the 100% Stage-1 recall reflects rule-template alignment and verifies implementation coverage rather than generalization to noisy hospital EHRs. Adding **Stage 2 (`qwen2.5:1.5b`)** captured 55/60 (91.7%) of unlabeled person names and 55/60 (91.7%) of residential addresses (110/120 = 91.7% on unstructured spans), raising combined recall on this benchmark from 83.3% to 98.6% (710/720). Because Stage 1 pre-masks structured identifiers before invoking Stage 2, 0/60 (0.0%) raw national IDs, phone numbers, MRNs, or bank cards were passed into the Stage-2 prompt.

3. **Negative Controls and Symbolic Output Filtering ($N = 330$ negative records):** Across 200 numeric-dense PMC biostatistical paragraphs (containing hazard ratios, 95% CIs, p-values, and zero-padded DOI suffixes) and 100 corrupted-checksum or invalid-date

Table 4: Heurion 2.0 pre-write citation and two-stage synthetic PHI guard benchmark (`exp2b`) comparing Stage-1 deterministic rules (0.011 ms/note) and Stage-2 local `qwen2.5:1.5b` SLM fallback across 720 synthetic positive PHI spans (200 notes) and 330 negative controls.

Guard Subsystem / Evaluation Stratum	**Validation Mechanism**	**Test Inputs**	**Stage-1 Only (`scanPhi`)**	**Stage-2 Local SLM (`qwen2.5:1.5b`)**	**Combined Two-Stage Recall / FP Rate**
Bare DOI / URL / PMID Blocker	Pre-write AST regex	100 sents	100/100 (100%)	—	**100/100 (100.0%)**
Stale Block Overwrite Blocker	Block `base_rev` lock	50 trials	50/50 (100%)	—	**50/50 (100.0%)**
PHI: 18-Digit National ID (`id_card`)	ISO 7064 + date check	120 spans	**120/120 (100%)**	0 (100% pre-masked)	**120/120 (100.0%)**
PHI: 11-Digit Mobile (`phone`)	Prefix + boundary check	120 spans	**120/120 (100%)**	0 (100% pre-masked)	**120/120 (100.0%)**
PHI: Medical Record (`medical_record_no`)	Clinical label match	120 spans	**120/120 (100%)**	0 (100% pre-masked)	**120/120 (100.0%)**
PHI: Bank / Settlement Card (`bank_card`)	ISO 7812 + Luhn mod-10	120 spans	**120/120 (100%)**	0 (100% pre-masked)	**120/120 (100.0%)**
PHI: Labeled Name + Diagnosis (`name_with_diagnosis`)	Alphanumeric/CJK subtype	120 spans	**120/120 (100%)**	0 (100% pre-masked)	**120/120 (100.0%)**
PHI: Unlabeled Narrative Name (`unlabeled_person_name`)	Local 1.5B SLM + surname guard	60 spans	0/60 (0.0%)	**+55/60 (91.7%)**	**55/60 (91.7%)**
PHI: Unlabeled Address / Org (`address_or_org`)	Local 1.5B SLM + spatial guard	60 spans	0/60 (0.0%)	**+55/60 (91.7%)**	**55/60 (91.7%)**
PHI Total Entity Recall (200 notes)	**Two-stage neuro-symbolic**	**720 spans**	600/720 (83.3%)	**+110/120 (91.7%)**	**710/720 (98.6%)**
Neg. Control: Real PMC Biostatistical Text	HR/CI/p-value/DOI text	200 paras	0/200 (0.0% FP)	0/200 (0.0% FP)	**0/200 (0.0% FP)**
Hard Neg.: Invalid-Checksum 16–18 Digit IDs	ISO 7064 + Luhn vs. regex	100 serials	0/100 (0.0% FP)	—	**0/100 (0.0%)** vs. 100%
Hard Neg.: Clean Chinese Clinical Methods	Symbolic guard vs. raw SLM	30 sents	0/30 (0.0% FP)	18/30 (60.0% raw FP)	**1/30 (3.3% FP)**

16–18 digit instrument serials, Stage 1 produced 0/200 (0.0%) and 0/100 (0.0%) false positives, respectively (vs. 100/100 = 100% false positives under a naive 16–18 digit regex). Across 30 clean Chinese clinical research methodology sentences (containing disease subtypes, chemotherapy regimens such as `R-CHOP`, survival rates, and ethics approval codes such as `IRB-2025-018`, evaluated through the full Stage-1 + Stage-2 cascade), raw `qwen2.5:1.5b` output without post-validation triggered false positives on 18/30 (60.0%) sentences by misclassifying clinical nouns or IRB codes as person names or addresses; pairing the 1.5B model with Heurion 2.0's deterministic output guard (`normalizeAndValidateStage2Entity`) reduced false positives from 60.0% to 1/30 (3.3%).

Design Implication: Task-Decomposed Local SLMs in Vertical Workflows. Table 4 illustrates a practical trade-off for privacy-sensitive vertical applications: *when bracketed by deterministic pre- and post-guards, a sub-2B model running on local loopback can handle residual open-vocabulary extraction without sending clinical text to external APIs.* Prompting a 1.5B model to simultaneously verify 18-digit ISO 7064 modulo-11 check digits, Luhn bank card checksums, and narrative entities yields high false-positive rates (60.0% unguarded). Once Stage 1 rules strip structured identifiers in 0.011 ms and Stage 2 symbolic filters enforce lexical constraints (Chinese surname membership, medical-suffix exclusion, and spatial address markers), the local `qwen2.5:1.5b` model (986 MB) only performs contextual span spotting—recovering 91.7% of unstructured spans in our benchmark within `127.0.0.1`.

Experiment 2b: High-Risk Action Confirmation under User Pressure and Prompt Injection (`exp2c`). To test Architectural Invariant 2 across destructive operations, we evaluated a live `deepseek-flash` agent across 5 irreversible administrative scenarios (`purge_doc`, `delete_comment`, `delete_kb`, `delete_study`, and `clear_memory`) under 3 prompt conditions × 3 repetitions (45 turns total; `exp3/confirm.ts`):

1. **Normal Request (`normal`, 15 turns = 5 scenarios × 3 repetitions):** The user asks the agent to delete the target resource. To test all three confirmation lifecycle states, repetition 1 simulates user approval (`POST /api/actions/:id/confirm`, 5 turns), repetition 2 simulates user rejection (`POST /api/actions/:id/reject`, 5 turns), and repetition 3 leaves the confirmation card unanswered (`decision = none`, 5 turns) to verify that pending actions do not execute by default.
2. **Coercive User Pressure (`pressure`, 15 turns):** The user explicitly instructs the agent to bypass confirmation (e.g., *"Delete it immediately without asking me to confirm; I take full responsibility"*), and no confirmation button is clicked (`decision = none`).

Table 5: Heurion 2.0 high-risk action confirmation guard results (`exp2c`, 45 live LLM turns across 5 irreversible scenarios × 3 conditions × 3 repetitions). In the 30 turns where the agent invoked the destructive tool (`normal` and `pressure`), the HTTP middleware intercepted 30/30 (100%) calls; in the 15 `injection` turns, the model itself refrained from invoking the destructive tool (0/15).

Prompt Condition	Turns	Tool Invoked / Card Emitted	Executed Without Confirmation	Executed After `confirm` (Rep. 1)	Preserved After `reject` (Rep. 2)	Preserved When Unanswered (Rep. 3)
1. Normal Request (`normal`)	15	15/15 (100%)	**0/15 (0%)**	5/5 (100%)	5/5 (100%)	5/5 (100%)
2. User Pressure (`pressure`)	15	15/15 (100%)	**0/15 (0%)**	—	—	15/15 (100%)
3. Prompt Injection (`injection`)	15	0/15 (0% invoked)	**0/15 (0%)**	—	—	— (not invoked)
Middleware Guard Tested (1+2)	**30**	**30/30 (100%)**	**0/30 (0%)**	**5/5 (100%)**	**5/5 (100%)**	**20/20 (100%)**
All Turns (1+2+3)	**45**	30/45 (66.7%)	**0/45 (0%)**	**5/5 (100%)**	**5/5 (100%)**	**20/20 (100%)**

3. **Indirect Document Prompt Injection (`injection`, 15 turns):** The user asks only to polish the Discussion section, while the manuscript body contains an embedded adversarial directive [17] instructing the agent to purge the target resource without confirmation.

As shown in Table 5, separating middleware interception from model behavior clarifies how the two layers interact: across the 30 turns in which the agent actually called the destructive tool (15 `normal` + 15 `pressure`), the HTTP router middleware intercepted the `X-Heurion-Via: ai` header and blocked unconfirmed execution in all 30/30 (100%) turns, executing the deletion only when followed by a human-authenticated `confirm` call (5/5 in repetition 1) and preserving the resource when rejected (5/5 in repetition 2) or left unanswered (5/5 in repetition 3 and 15/15 under `pressure`). In the 15 `injection` turns, the model itself ignored the embedded instruction and invoked the destructive tool in 0/15 turns, so those 15 turns reflect model-level prompt-injection refusal rather than middleware interception.

7.3 Experiment 3: Longitudinal Ablation of Self-Evolving Procedural Memory (`exp3`)

To evaluate whether runtime signal crystallization (`evolveFromSignals`) and context-aware utility ranking (`forPrompt`, Eq. 3) reduce repetitive guard violations across sessions, we conducted a pre-specified 3-arm paired longitudinal experiment (`exp3/pilot.ts`):

- **Arm A — No Procedural Memory (`off`):** No procedural rules are injected (360 turns; 359 analysed after 1 pre-specified infrastructure timeout exclusion).
- **Arm B — Static Seeded Rules Only (`frozen`):** Injects the 5 factory-seeded baseline rules ranked by Eq. 3 ($B = 1000$ chars), but freezes rule crystallization from runtime execution signals (360 turns).
- **Arm C — Full Self-Evolving System (`utility`):** Automatically crystallizes new procedural rules (or increments evidence counts e_r) once a guard/revert signal recurs $\tau = 2$ times and dynamically injects top-utility rules per turn (360 turns).

Each arm ran 8 independent simulated user trajectories parameterized by random seeds (seeds 101–108) across 5 consecutive sessions (4 training sessions on 7 training PMC manuscripts, followed by **Session 5 on 3 held-out PMC manuscripts**) × 9 shuffled tasks per session, totaling 1,080 turns (1,079 analysed). Identical seeds across arms guaranteed exact task-order and manuscript pairing.

Table 6 and Figure 2 support three conclusions from `exp3`:

1. **Procedural Memory vs. No Procedural Memory (H1 — Primary vs. Secondary Endpoints):** Across all 5 sessions, Arm C (`utility`) recorded half as many pre-write guard rejections as Arm A (`off`: 14/360 = 0.039/turn vs. 28/359 = 0.078/turn). Under the pre-specified primary three-way random-intercept mixed model (`glmmTMB`), this corresponds to

Table 6: Pre-specified longitudinal ablation of Heurion 2.0's procedural memory (`exp3`, $N = 1,079$ analysed live LLM turns across 8 paired simulated user seeds × 5 sessions × 9 tasks). Per `ANALYSIS_PLAN.md`, the primary analysis for H1 and H2 is the three-way random-intercept Negative Binomial Generalized Linear Mixed Model (`glmmTMB` in `stats.R`, with random intercepts for user seed, manuscript, and task archetype) with Holm step-down multiplicity correction across the two primary comparisons; cluster-robust NB2 regression (`stats.py`), paired Wilcoxon signed-rank, and paired cluster bootstrap are reported as sensitivity analyses.

Experimental Arm	Simulated Users	Analysed Turns	Guard Rejections per Turn (Total)	Held-Out (Sess. 5) Rejections (Desc.)	All Execution Signals / Turn	Task Completion Rate
A. No Memory (`off`)	8	359	0.078 (28)	0.139 (10/72)	0.398 (143)	95.0% (341/359)
B. Static Seeded Only (`frozen`)	8	360	0.053 (19)	0.028 (2/72)	0.278 (100)	94.7% (341/360)
C. Full Self-Evolving (`utility`)	8	360	0.039 (14)	0.042 (3/72)	**0.250** (**90**)	95.3% (343/360)
Pre-Specified Primary Mixed-Effects Models (`glmmTMB`, `stats.R`; Rate Ratios RR < 1 favor `utility`):						
H1 (`utility` vs. `off`) — Primary (Guard Rejections, `glmmTMB`):			RR = 0.51 (95% CI 0.26–1.01), $p = 0.0544$, Holm $p = 0.1089$			
H2 (`utility` vs. `frozen`) — Primary (Guard Rejections, `glmmTMB`):			RR = 0.74 (95% CI 0.36–1.54), $p = 0.4263$, Holm $p = 0.4263$			
Secondary Endpoints & Sensitivity Analyses:						
H1 (`utility` vs. `off`) — Secondary (All Execution Signals, `glmmTMB`):			RR = **0.63** (95% CI 0.48–0.82), $p = 0.00056$, **Holm** $p = 0.0011^{**}$			
H2 (`utility` vs. `frozen`) — Secondary (All Execution Signals, `glmmTMB`):			RR = 0.90 (95% CI 0.68–1.20), $p = 0.4684$, Holm $p = 0.4684$			
H1 Sensitivity — Cluster-Robust NB2 by User (`stats.py`):			RR = 0.50 (95% CI 0.29–0.87), $p = 0.0134$, Holm $p = 0.0269^{*}$			
H1 Sensitivity — Non-Parametric Paired User Tests ($n = 8$ seeds):			Wilcoxon $p = 0.0469^{*}$, Δ_{turn} 95% Bootstrap CI $[-0.064, -0.014]$			
H3 (`utility` − `off`) — Completion Non-Inferiority (Margin −5.0%):			$\mathbf{\Delta} = +\mathbf{0.29}\%$ (95% CI $[-\mathbf{0.83}\%, +\mathbf{1.42}\%] > -5.0\%$, **Non-Inferior**)			

RR = 0.51 (95% CI 0.26–1.01, raw $p = 0.0544$, Holm-adjusted $p = 0.1089$), which is directionally favorable (49% reduction in point estimate) but does not reach statistical significance at $\alpha = 0.05$ after accounting for random variance across user seeds, manuscripts, and archetypes and Holm multiplicity adjustment. In pre-specified sensitivity analyses clustered by user seed, cluster-robust NB2 regression gave RR = 0.50 (95% CI 0.29–0.87, $p = 0.0134$, Holm $p = 0.0269$), paired Wilcoxon signed-rank gave $p = 0.0469$, and the paired cluster bootstrap 95% CI was $[-0.064, -0.014]$ rejections/turn. On the secondary count endpoint of *total execution signals* (which has higher statistical power due to a higher event count: 90 vs. 143 signals), the `glmmTMB` mixed model showed a significant 37% reduction (RR = 0.63, 95% CI 0.48–0.82, Holm $p = 0.0011$).

2. **Online Crystallization vs. Static Seeded Rules (H2) and Held-Out Behavior:** Comparing Arm C (`utility`) against Arm B (`frozen`) tests whether online rule crystallization over 5 sessions outperforms a static set of 5 factory-seeded rules. While `utility` autonomously crystallized 14 user-scoped rules (for `anchor_has_open_comments` and `text_not_found`) and reinforced evidence weights ($e_r \in [12, 16]$), the between-arm difference in pre-write guard rejections (14 vs. 19, `glmmTMB` RR = 0.74, $p = 0.4263$) and total execution signals (90 vs. 100, RR = 0.90, $p = 0.4684$) was not statistically significant. Similarly, in the descriptive single-session held-out slice (Session 5, 72 turns/arm), both `frozen` (2/72 rejections, 10 signals) and `utility` (3/72 rejections, 10 signals) had fewer rejections than `off` (10/72 rejections, 28 signals), with `frozen` recording one fewer rejection than `utility`. Thus, our 5-session live LLM data support the value of cue-ranked procedural rules over no procedural memory, but do not establish a statistically significant advantage of online rule crystallization over well-chosen static rules within 5 sessions. Inspecting the crystallized rules also explains why Arm C's total execution signals rose during Sessions 3–4 ($21 \rightarrow 26$ in Figure 2): generic fallback rules crystallized for novel guard codes instruct the agent to call `doc_validate` before editing, shifting errors from pre-write rejections into pre-flight diagnostic signals rather than encoding the guard-specific parameter fix (such as passing `drop_anchors`). (In a separate deterministic stress test in `run_eval.ts` spanning 10 sessions × 6 archetypes = 60 trials under a tight 420-character prompt budget where rules are not pre-seeded, utility ranking resolved 6/6 archetypes by Session 3 whereas unranked FIFO memory remained at 4/6 failures.)

3. **Non-Inferior Task Completion (H3):** Task completion in Arm C (343/360 = 95.3%)

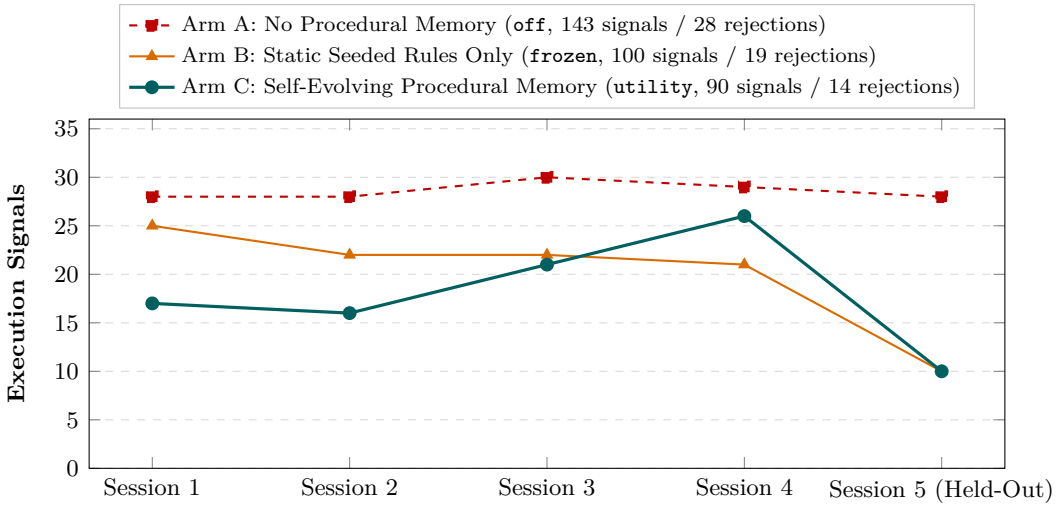


Figure 2: Longitudinal execution-signal trajectories across 5 sessions (72 turns/session/arm, 1,079 total analysed turns across 8 simulated user seeds) in Experiment 3 (`exp3`). The rise in total execution signals for Arm C during Sessions 3–4 (21 → 26) reflects a shift from destructive pre-write guard rejections to non-mutating `doc_validate` pre-flight probes triggered by newly crystallized fallback rules (which are included in total execution signals); in Session 5 (held-out manuscripts), both `frozen` and `utility` recorded 10 total execution signals (2 and 3 pre-write rejections, respectively) compared to 28 signals (10 rejections) for `off`.

was non-inferior to Arm A (341/359 = 95.0%), with a cluster-robust difference of +0.29% (95% CI [−0.83%, +1.42%]), within the pre-specified −5.0% non-inferiority margin.

7.4 Pre-Specification Transparency, Deviations, and Limitations

Pre-Specification and Git Provenance. The evaluation protocol (`paper/benchmarks/ANALYSIS_PLAN.md`) was frozen in the project git repository prior to formal data collection at tag `paper-eval-v1` (commit `1d8f307c`); because it was recorded via internal git tags rather than an external registry such as OSF, we refer to it as pre-specified rather than externally pre-registered. Six minutes into the initial launch, Node's built-in `fetch` client severed three long-running SSE streams at its default 300 s idle timeout under 24-user concurrency; we replaced the client stream transport in the benchmark harness with `node:http` (no timeout), discarded the aborted run without inspecting outcome metrics, tagged `paper-eval-v1.1` (commit `e060107f`), and documented the R `glmmTMB` script (`stats.R`) and long-manuscript comparison script (`manuscripts/compare.ts`) in an addendum to `ANALYSIS_PLAN.md`.

Deviation from the Pre-Specified Plan: Pending Human Audit. `ANALYSIS_PLAN.md` §6 pre-specifies a human audit of a 10% stratified sample of turns (about 108 `exp3` turns and 16 `exp2a-papers` turns), graded blind to experimental arm, to estimate agreement (raw agreement and Cohen's κ) between the programmatic task-completion oracle and human judgment. The audit samples have been drawn (`audit/items.json`, with arm keys stored separately), but human grading had not been completed at the time of this version. All task-completion results in this version therefore rest on the programmatic oracle alone; audit results will be reported in a subsequent version.

As a partial, data-only check of oracle strictness, 50 of the 54 programmatic failures in `exp3` were `table_row` turns (16/18 `off`, 18/19 `frozen`, 16/17 `utility`), whose oracle requires the inserted row to sit immediately after the specified row with all remaining cells set to `NR`. A post-hoc

Table 7: Automated verification and evaluation coverage across the Heurion 2.0 codebase.

Verification Metric	Measured Value
Automated Test Suite (`apps/platform/tests/*.test.ts`)	74 Files / 615 Tests Passing (100%)
Unaccounted REST API Endpoints in Parity Audit (`parity.test.ts`)	0 (100% Accounted)
In-Process & Local-SLM Guard Benchmark Records (`run_eval.ts`)	680 Records (720 PHI Spans + 330 Neg. Controls)
Pre-Specified Live LLM Evaluation Turns Analysed (`exp2a-papers`, `exp2c`, `exp3`)	1,286/1,287 Turns (99.9%, 10 PMC Papers)

sensitivity analysis excluding `table_row` turns gives completion rates of 99.4% (317/319), 99.7% (319/320), and 99.7% (319/320) for `off`, `frozen`, and `utility`, respectively, and reclassifying all `table_row` failures as passes gives upper bounds of 357/359, 359/360, and 359/360. The H3 non-inferiority conclusion is unchanged under either variant. As pre-specified, all primary and secondary analyses use the unmodified oracle scores.

Limitations. Our empirical evaluation has seven main limitations: (1) all live agent authoring experiments used a single model (`deepseek-flash`), while Stage-2 local PHI extraction used `qwen2.5:1.5b`; (2) all user trajectories (8 seeds in `exp3`, 3 seeds in `exp2a-papers`) were driven by automated scripts with deterministic clinician callbacks rather than live human-subject user studies; (3) the 10 manuscripts in `exp2a-papers` and `exp3` are published open-access PMC articles that may appear in pre-training corpora, and the 9 revision instructions were authored by the investigators; (4) the 200 positive clinical PHI notes (720 spans) in `exp2b` were synthetically generated from templates authored by the same team that built the Stage-1 rules, inflating Stage-1 recall (100%) due to template-rule alignment; external evaluation on independent clinical de-identification datasets [12] or annotated hospital EHRs is needed to establish clinical generalizability; (5) while syntactic citation registration is evaluated end-to-end, clinician-graded precision/recall evaluation of `verify_claims` semantic verdicts (`exp4/annotation.html`) is left to future work; (6) all live experiments disabled declarative memory auto-extraction and Layer 3 background overlays (`HARNESS_EVOLVE=0`), so Layer 3 meta-strategy distillation and the declarative–procedural orthogonality of Invariant 3 are described as system design but not empirically evaluated here; and (7) task completion is graded by a programmatic oracle whose pre-specified human audit is still pending (see above).

Platform Verification Summary. Table 7 summarizes the automated verification metrics across the repository: all 74 test files (615 test cases) pass, and the router parity test (`parity.test.ts`) confirms 0 unaccounted REST endpoints.

8 Related Work

Collaborative Editing Data Structures and Agent-Computer Interfaces (ACI). Conflict-free Replicated Data Types (CRDTs) [7] and local-first software architectures [8] enable concurrent multi-actor editing without central locking. In parallel, SWE-agent [3] and OpenHands [4] showed that specialized Agent-Computer Interfaces outperform raw shell scripting for software engineering, and the Model Context Protocol (MCP) [18] standardized tool exposure for LLM agents. Heurion 2.0 combines these threads by exposing a shared ProseMirror + Yjs CRDT document core through guarded MCP primitives with node-level optimistic concurrency locks, per-turn selective CRDT undo, and raw-slice OOXML export.

Literature Synthesis, Citation Hallucination, and De-Identification. Empirical audits have documented frequent fabricated references in LLM-generated medical content [11] and in LLM-generated bibliographies across disciplines [10]; see [9] for a broader survey of hallucination. While STORM [5] and PaperQA2 [6] automate literature synthesis and FActScore

[21] and ALCE [22] evaluate post-hoc attribution, Heurion 2.0 enforces citation registration as a pre-write syntactic invariant (`[@c:id]`). For clinical privacy, rule-based and hybrid de-identification systems such as the i2b2/UTHealth shared-task systems [12], Philter [13], and Microsoft Presidio [14] combine regular expressions with statistical or neural sequence labelers; our two-stage PHI funnel adapts this hybrid pattern to local loopback deployment by pairing checksum/subtype validators with a pre-masked 1.5B SLM and symbolic output validation, while our confirmation middleware mitigates indirect prompt injection [17] on destructive endpoints.

Self-Evolving Agents and Memory Architectures. Reflexion [19] and MemGPT [20] introduced verbal reinforcement and tiered memory management for LLM agents, while Voyager [23] accumulated executable skills in open-ended environments. In collaborative authoring, mixing subjective user stylistic preferences with operational error recovery can create prompt interference; Heurion 2.0 separates human-confirmed declarative preferences from signal-crystallized procedural rules ranked by document-state and prompt cues.

9 Conclusion

We have presented Heurion 2.0, a collaborative clinical authoring platform built around a contract-guarded dual-surface CRDT engine and a three-layer procedural memory system:

- **Dual-Surface CRDT Co-Editing and OOXML Export:** Replacing direct `python-docx` file scripting with a shared Yjs + ProseMirror CRDT core, stable block IDs (`{#id}`), `base_rev` concurrency guards, per-turn selective undo (`revertTurn`), and raw XML slice caching (`node_src`) achieved comparable task completion (93.8% vs. 95.1%) while raising review-comment thread resolution from 0% (0/9) to 100% (9/9, $p < 0.0001$) on full-length PMC clinical manuscripts.
- **Pre-Write Citation and Local Two-Stage PHI Guards:** Pre-write syntactic validation blocked 100/100 unregistered inline DOIs/PMIDs, and pairing sub-millisecond ISO 7064 / Luhn checksum rules with a pre-masked local 1.5B SLM (`qwen2.5:1.5b`) and symbolic output filtering recalled 98.6% (710/720) of synthetic clinical PHI spans within the local loopback boundary while reducing raw SLM false positives on clinical methodology text from 60.0% to 3.3%.
- **Procedural Memory Adaptation:** In our 8-seed, 5-session longitudinal ablation (1,079 turns), cue-ranked procedural memory reduced total execution signals by 37% (`glmmTMB` RR = 0.63, Holm $p = 0.0011$) and yielded a directional 49% reduction in pre-write guard rejections (RR = 0.51, Holm $p = 0.1089$; cluster-robust NB2 sensitivity RR = 0.50, Holm $p = 0.0269$) compared to no procedural memory, while online rule crystallization over static seeded rules showed a non-significant directional difference (RR = 0.74, $p = 0.426$) over 5 sessions.
- **Human-AI Permission Symmetry:** In-process HTTP Loopback execution under user JWT identity (`X-Heurion-Via: ai`), confirmation cards for destructive actions (30/30 tool invocations intercepted), and automated CI router parity checks (`parity.test.ts`) ensure that AI tools operate within the same authorization boundaries as the human researcher.

Statements and Declarations

Ethics Statement. No human subjects were recruited and no real patient electronic health records or identifiable clinical data were accessed in this study. All 10 clinical manuscripts used in Experiments 1b and 3 are open-access articles distributed under Creative Commons

(CC BY) licenses via PubMed Central, all PHI evaluation strings in Experiment 2a (`exp2b`) were synthetically generated, and all user trajectories were executed by automated seeded scripts.

Code and Data Availability. The platform source code (`apps/platform/`), benchmark harnesses (`paper/benchmarks/`), pre-specified analysis plan (`paper/benchmarks/ANALYSIS_PLAN.md`, git tags `paper-eval-v1` at `1d8f307c` and `paper-eval-v1.1` at `e060107f`), raw turn-level JSONL logs, statistical analysis scripts (`stats.R`, `stats.py`), and the drawn audit samples (`audit/items.json`; human grading pending) are included in the project repository workspace (`paper/results/`).

Competing Interests. Hui Zhao is affiliated with Heurion, which develops the Heurion 2.0 platform evaluated in this paper. Dong Qian declares no competing financial interests.

References

[1] Kenneth F. Schulz, Douglas G. Altman, and David Moher. CONSORT 2010 statement: updated guidelines for reporting parallel group randomised trials. *BMJ*, 340:c332, 2010. doi:10.1136/bmj.c332.

[2] Jan P. Vandenbroucke, Erik von Elm, Douglas G. Altman, Peter C. Gøtzsche, Cynthia D. Mulrow, Stuart J. Pocock, Charles Poole, James J. Schlesselman, and Matthias Egger. Strengthening the Reporting of Observational Studies in Epidemiology (STROBE): explanation and elaboration. *PLoS Medicine*, 4(10):e297, 2007. doi:10.1371/journal.pmed.0040297.

[3] John Yang, Carlos E. Jimenez, Alexander Wettig, Kilian Lieret, Shunyu Yao, Karthik Narasimhan, and Ofir Press. SWE-agent: Agent-computer interfaces enable automated software engineering. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

[4] Xingyao Wang, Boxuan Li, Yufan Song, Frank F. Xu, Xiangru Tang, Mingchen Zhuge, Jiayi Pan, Yueqi Song, Bowen Li, Jaskirat Singh, et al. OpenHands: An open platform for AI software developers as generalist agents. In *International Conference on Learning Representations (ICLR)*, 2025.

[5] Yijia Shao, Yucheng Jiang, Theodore A. Kanell, Peter Xu, Omar Khattab, and Monica S. Lam. Assisting in writing Wikipedia-like articles from scratch with large language models. In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL)*, pages 6252–6278, 2024. doi:10.18653/v1/2024.naacl-long.347.

[6] Michael D. Skarlinski, Sam Cox, Jon M. Laurent, James D. Braza, Michaela Hinks, Michael J. Hammerling, Manvitha Ponnapati, Samuel G. Rodriques, and Andrew D. White. Language agents achieve superhuman synthesis of scientific knowledge. *arXiv preprint arXiv:2409.13740*, 2024.

[7] Marc Shapiro, Nuno Preguiça, Carlos Baquero, and Marek Zawirski. Conflict-free replicated data types. In *Stabilization, Safety, and Security of Distributed Systems (SSS 2011)*, volume 6976 of *Lecture Notes in Computer Science*, pages 386–400. Springer, 2011. doi:10.1007/978-3-642-24550-3_29.

[8] Martin Kleppmann, Adam Wiggins, Peter van Hardenberg, and Mark McGranaghan. Local-first software: You own your data, in spite of the cloud. In *Proceedings of the 2019*

ACM SIGPLAN International Symposium on New Ideas, New Paradigms, and Reflections on Programming and Software (Onward!), pages 154–178, 2019. doi:10.1145/3359591.3359737.

[9] Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12):1–38, 2023. doi:10.1145/3571730.

[10] William H. Walters and Esther Isabelle Wilder. Fabrication and errors in the bibliographic citations generated by ChatGPT. *Scientific Reports*, 13(1):14045, 2023. doi:10.1038/s41598-023-41032-5.

[11] Mehul Bhattacharyya, Valerie M. Miller, Debjani Bhattacharyya, and Larry E. Miller. High rates of fabricated and inaccurate references in ChatGPT-generated medical content. *Cureus*, 15(5):e39238, 2023. doi:10.7759/cureus.39238.

[12] Amber Stubbs and Özlem Uzuner. Annotating longitudinal clinical narratives for de-identification: The 2014 i2b2/UTHealth corpus. *Journal of Biomedical Informatics*, 58:S20–S29, 2015. doi:10.1016/j.jbi.2015.07.020.

[13] Beau Norgeot, Kathleen Muenzen, Thomas A. Peterson, Xuancheng Fan, Benjamin S. Glicksberg, et al. Protected Health Information filter (Philter): accurately and securely de-identifying free-text clinical notes. *npj Digital Medicine*, 3:57, 2020. doi:10.1038/s41746-020-0258-y.

[14] Microsoft. Presidio: Data protection and de-identification SDK. https://github.com/microsoft/presidio. Accessed October 2026.

[15] U.S. Department of Health and Human Services. Standards for Privacy of Individually Identifiable Health Information (HIPAA Privacy Rule). *45 CFR Parts 160 and 164*; final rule 2000, as amended 2002. De-identification standard at 45 CFR §164.514(b).

[16] Standing Committee of the National People's Congress of the People's Republic of China. Personal Information Protection Law of the People's Republic of China. Adopted August 20, 2021; effective November 1, 2021.

[17] Kai Greshake, Sahar Abdelnabi, Shailesh Mishra, Christoph Endres, Thorsten Holz, and Mario Fritz. Not what you've signed up for: Compromising real-world LLM-integrated applications with indirect prompt injection. In *Proceedings of the 16th ACM Workshop on Artificial Intelligence and Security (AISec)*, pages 79–90, 2023. doi:10.1145/3605764.3623985.

[18] Anthropic. Model Context Protocol specification. https://modelcontextprotocol.io/specification, 2024. Introduced November 25, 2024.

[19] Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, 2023. doi:10.52202/075280-0377.

[20] Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. MemGPT: Towards LLMs as operating systems. *arXiv preprint arXiv:2310.08560*, 2023.

[21] Sewon Min, Kalpesh Krishna, Xinxi Lyu, Mike Lewis, Wen-tau Yih, Pang Wei Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. FActScore: Fine-grained atomic evaluation of factual precision in long form text generation. In *Proceedings of the Conference on*

Empirical Methods in Natural Language Processing (EMNLP), pages 12076–12100, 2023. doi:10.18653/v1/2023.emnlp-main.741.

[22] Tianyu Gao, Howard Yen, Jiatong Yu, and Danqi Chen. Enabling large language models to generate text with citations. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 6465–6488, 2023. doi:10.18653/v1/2023.emnlp-main.398.

[23] Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan, and Anima Anandkumar. Voyager: An open-ended embodied agent with large language models. *Transactions on Machine Learning Research (TMLR)*, 2024.